



MATH240: First Order Separable

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Separable ODEs

1. Solve the separable ODE: $dy/dx = 6x^1$. Find the general solution $y(x)$.

$$\int 6x^1 dx$$

Answer: _____

2. A particle's velocity satisfies $dv/dt = 1t^2$. Find the position function (antiderivative).

$$\int 1x^2 dx$$

Answer: _____

3. Solve: $dy/dx = 5x^3$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 5x^3 dx$$

Answer: _____

4. Solve: $dy/dx = 5x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 5x^3 dx$$

Answer: _____

5. Solve the separable ODE: $dy/dx = 1x^4$. Find the general solution $y(x)$.

$$\int 1x^4 dx$$

Answer: _____



6. A particle's velocity satisfies $dv/dt = 3t^1$. Find the position function (antiderivative).

$$\int 3x^1 dx$$

Answer: _____

7. Solve: $dy/dx = 6x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 6x^2 dx$$

Answer: _____

8. Solve: $dy/dx = 2x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 2x^3 dx$$

Answer: _____

9. Solve the separable ODE: $dy/dx = 4x^3$. Find the general solution $y(x)$.

$$\int 4x^3 dx$$

Answer: _____

10. A particle's velocity satisfies $dv/dt = 2t^3$. Find the position function (antiderivative).

$$\int 2x^3 dx$$

Answer: _____

11. Solve: $dy/dx = 4x^1$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 4x^1 dx$$

Answer: _____

12. Solve: $dy/dx = 4x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 4x^3 dx$$

Answer: _____

13. Solve the separable ODE: $dy/dx = 2x^4$. Find the general solution $y(x)$.

$$\int 2x^4 dx$$

Answer: _____



14. A particle's velocity satisfies $dv/dt = 1t^1$. Find the position function (antiderivative).

$$\int 1x^1 dx$$

Answer: _____

15. Solve: $dy/dx = 8x^3$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 8x^3 dx$$

Answer: _____

16. Solve: $dy/dx = 1x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 1x^3 dx$$

Answer: _____

17. Solve the separable ODE: $dy/dx = 5x^2$. Find the general solution $y(x)$.

$$\int 5x^2 dx$$

Answer: _____

18. A particle's velocity satisfies $dv/dt = 2t^1$. Find the position function (antiderivative).

$$\int 2x^1 dx$$

Answer: _____

19. Solve: $dy/dx = 4x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 4x^2 dx$$

Answer: _____

20. Solve: $dy/dx = 8x^2$. Find the coefficient of the x^3 term in the general solution.

$$\int 8x^2 dx$$

Answer: _____

21. Solve the separable ODE: $dy/dx = 5x^2$. Find the general solution $y(x)$.

$$\int 5x^2 dx$$

Answer: _____



22. A particle's velocity satisfies $dv/dt = 4t^2$. Find the position function (antiderivative).

$$\int 4x^2 dx$$

Answer: _____

23. Solve: $dy/dx = 5x^1$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 5x^1 dx$$

Answer: _____

24. Solve: $dy/dx = 7x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 7x^3 dx$$

Answer: _____

25. Solve the separable ODE: $dy/dx = 1x^3$. Find the general solution $y(x)$.

$$\int 1x^3 dx$$

Answer: _____

26. A particle's velocity satisfies $dv/dt = 1t^1$. Find the position function (antiderivative).

$$\int 1x^1 dx$$

Answer: _____

27. Solve: $dy/dx = 7x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 7x^2 dx$$

Answer: _____

28. Solve: $dy/dx = 1x^2$. Find the coefficient of the x^3 term in the general solution.

$$\int 1x^2 dx$$

Answer: _____

29. Solve the separable ODE: $dy/dx = 1x^3$. Find the general solution $y(x)$.

$$\int 1x^3 dx$$

Answer: _____



30. A particle's velocity satisfies $dv/dt = 3t^3$. Find the position function (antiderivative).

$$\int 3x^3 dx$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Separable ODEs. All answers verified by independent computation.

Solutions



Separable ODEs

1. Solve the separable ODE: $dy/dx = 6x^1$. Find the general solution $y(x)$.

$$\int 6x^1 dx$$

→ Separate variables: $dy = 6x^1 dx$.

→ Integrate both sides: $y = \text{integral of } 6x^1 dx = 3x^2 + C$.

Answer: $\frac{6}{2}x^2 + C \Rightarrow \text{coeff} = 3$

2. A particle's velocity satisfies $dv/dt = 1t^2$. Find the position function (antiderivative).

$$\int 1x^2 dx$$

→ Integrate: $v(t) = \text{integral of } 1t^2 dt = 1/3t^3 + C$.

Answer: $\frac{1}{3}x^3 + C \Rightarrow \text{coeff} = 1/3$

3. Solve: $dy/dx = 5x^3$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 5x^3 dx$$

→ Integrate: $y = 5/4x^4 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 5/4x^4$.

Answer: $\frac{5}{4}x^4 + C \Rightarrow \text{coeff} = 5/4$

4. Solve: $dy/dx = 5x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 5x^3 dx$$

→ Separate and integrate: $y = 5/4x^4 + C$.

→ The coefficient of x^4 is $5/4$.

Answer: $\frac{5}{4}x^4 + C \Rightarrow \text{coeff} = 5/4$

5. Solve the separable ODE: $dy/dx = 1x^4$. Find the general solution $y(x)$.

$$\int 1x^4 dx$$

→ Separate variables: $dy = 1x^4 dx$.

→ Integrate both sides: $y = \text{integral of } 1x^4 dx = 1/5x^5 + C$.

Answer: $\frac{1}{5}x^5 + C \Rightarrow \text{coeff} = 1/5$

6. A particle's velocity satisfies $dv/dt = 3t^1$. Find the position function (antiderivative).

$$\int 3x^1 dx$$

→ Integrate: $v(t) = \text{integral of } 3t^1 dt = 3/2t^2 + C$.

Answer: $\frac{3}{2}x^2 + C \Rightarrow \text{coeff} = 3/2$



7. Solve: $dy/dx = 6x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 6x^2 dx$$

→ Integrate: $y = 2x^3 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 2x^3$.

Answer: $\frac{6}{3}x^3 + C \Rightarrow \text{coeff} = 2$

8. Solve: $dy/dx = 2x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 2x^3 dx$$

→ Separate and integrate: $y = 1/2x^4 + C$.

→ The coefficient of x^4 is $1/2$.

Answer: $\frac{2}{4}x^4 + C \Rightarrow \text{coeff} = 1/2$

9. Solve the separable ODE: $dy/dx = 4x^3$. Find the general solution $y(x)$.

$$\int 4x^3 dx$$

→ Separate variables: $dy = 4x^3 dx$.

→ Integrate both sides: $y = \text{integral of } 4x^3 dx = 1x^4 + C$.

Answer: $\frac{4}{4}x^4 + C \Rightarrow \text{coeff} = 1$

10. A particle's velocity satisfies $dv/dt = 2t^3$. Find the position function (antiderivative).

$$\int 2x^3 dx$$

→ Integrate: $v(t) = \text{integral of } 2t^3 dt = 1/2t^4 + C$.

Answer: $\frac{2}{4}x^4 + C \Rightarrow \text{coeff} = 1/2$

11. Solve: $dy/dx = 4x^1$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 4x^1 dx$$

→ Integrate: $y = 2x^2 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 2x^2$.

Answer: $\frac{4}{2}x^2 + C \Rightarrow \text{coeff} = 2$

12. Solve: $dy/dx = 4x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 4x^3 dx$$

→ Separate and integrate: $y = 1x^4 + C$.

→ The coefficient of x^4 is 1 .

Answer: $\frac{4}{4}x^4 + C \Rightarrow \text{coeff} = 1$



13. Solve the separable ODE: $dy/dx = 2x^4$. Find the general solution $y(x)$.

$$\int 2x^4 dx$$

→ Separate variables: $dy = 2x^4 dx$.

→ Integrate both sides: $y = \text{integral of } 2x^4 dx = 2/5x^5 + C$.

Answer: $\frac{2}{5}x^5 + C \Rightarrow \text{coeff} = 2/5$

14. A particle's velocity satisfies $dv/dt = 1t^1$. Find the position function (antiderivative).

$$\int 1x^1 dx$$

→ Integrate: $v(t) = \text{integral of } 1t^1 dt = 1/2t^2 + C$.

Answer: $\frac{1}{2}x^2 + C \Rightarrow \text{coeff} = 1/2$

15. Solve: $dy/dx = 8x^3$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 8x^3 dx$$

→ Integrate: $y = 2x^4 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 2x^4$.

Answer: $\frac{8}{4}x^4 + C \Rightarrow \text{coeff} = 2$

16. Solve: $dy/dx = 1x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 1x^3 dx$$

→ Separate and integrate: $y = 1/4x^4 + C$.

→ The coefficient of x^4 is $1/4$.

Answer: $\frac{1}{4}x^4 + C \Rightarrow \text{coeff} = 1/4$

17. Solve the separable ODE: $dy/dx = 5x^2$. Find the general solution $y(x)$.

$$\int 5x^2 dx$$

→ Separate variables: $dy = 5x^2 dx$.

→ Integrate both sides: $y = \text{integral of } 5x^2 dx = 5/3x^3 + C$.

Answer: $\frac{5}{3}x^3 + C \Rightarrow \text{coeff} = 5/3$

18. A particle's velocity satisfies $dv/dt = 2t^1$. Find the position function (antiderivative).

$$\int 2x^1 dx$$

→ Integrate: $v(t) = \text{integral of } 2t^1 dt = 1t^2 + C$.

Answer: $\frac{2}{2}x^2 + C \Rightarrow \text{coeff} = 1$



19. Solve: $dy/dx = 4x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 4x^2 dx$$

→ Integrate: $y = 4/3x^3 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 4/3x^3$.

Answer: $\frac{4}{3}x^3 + C \Rightarrow \text{coeff} = 4/3$

20. Solve: $dy/dx = 8x^2$. Find the coefficient of the x^3 term in the general solution.

$$\int 8x^2 dx$$

→ Separate and integrate: $y = 8/3x^3 + C$.

→ The coefficient of x^3 is $8/3$.

Answer: $\frac{8}{3}x^3 + C \Rightarrow \text{coeff} = 8/3$

21. Solve the separable ODE: $dy/dx = 5x^2$. Find the general solution $y(x)$.

$$\int 5x^2 dx$$

→ Separate variables: $dy = 5x^2 dx$.

→ Integrate both sides: $y = \text{integral of } 5x^2 dx = 5/3x^3 + C$.

Answer: $\frac{5}{3}x^3 + C \Rightarrow \text{coeff} = 5/3$

22. A particle's velocity satisfies $dv/dt = 4t^2$. Find the position function (antiderivative).

$$\int 4x^2 dx$$

→ Integrate: $v(t) = \text{integral of } 4t^2 dt = 4/3t^3 + C$.

Answer: $\frac{4}{3}x^3 + C \Rightarrow \text{coeff} = 4/3$

23. Solve: $dy/dx = 5x^1$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 5x^1 dx$$

→ Integrate: $y = 5/2x^2 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 5/2x^2$.

Answer: $\frac{5}{2}x^2 + C \Rightarrow \text{coeff} = 5/2$

24. Solve: $dy/dx = 7x^3$. Find the coefficient of the x^4 term in the general solution.

$$\int 7x^3 dx$$

→ Separate and integrate: $y = 7/4x^4 + C$.

→ The coefficient of x^4 is $7/4$.

Answer: $\frac{7}{4}x^4 + C \Rightarrow \text{coeff} = 7/4$



25. Solve the separable ODE: $dy/dx = 1x^3$. Find the general solution $y(x)$.

$$\int 1x^3 dx$$

→ Separate variables: $dy = 1x^3 dx$.

→ Integrate both sides: $y = \text{integral of } 1x^3 dx = 1/4x^4 + C$.

Answer: $\frac{1}{4}x^4 + C \Rightarrow \text{coeff} = 1/4$

26. A particle's velocity satisfies $dv/dt = 1t^1$. Find the position function (antiderivative).

$$\int 1x^1 dx$$

→ Integrate: $v(t) = \text{integral of } 1t^1 dt = 1/2t^2 + C$.

Answer: $\frac{1}{2}x^2 + C \Rightarrow \text{coeff} = 1/2$

27. Solve: $dy/dx = 7x^2$. Then find the particular solution satisfying $y(0) = 0$.

$$\int 7x^2 dx$$

→ Integrate: $y = 7/3x^3 + C$.

→ Apply $y(0)=0$: $0 = 0 + C$, so $C = 0$.

→ Particular solution: $y = 7/3x^3$.

Answer: $\frac{7}{3}x^3 + C \Rightarrow \text{coeff} = 7/3$

28. Solve: $dy/dx = 1x^2$. Find the coefficient of the x^3 term in the general solution.

$$\int 1x^2 dx$$

→ Separate and integrate: $y = 1/3x^3 + C$.

→ The coefficient of x^3 is $1/3$.

Answer: $\frac{1}{3}x^3 + C \Rightarrow \text{coeff} = 1/3$

29. Solve the separable ODE: $dy/dx = 1x^3$. Find the general solution $y(x)$.

$$\int 1x^3 dx$$

→ Separate variables: $dy = 1x^3 dx$.

→ Integrate both sides: $y = \text{integral of } 1x^3 dx = 1/4x^4 + C$.

Answer: $\frac{1}{4}x^4 + C \Rightarrow \text{coeff} = 1/4$

30. A particle's velocity satisfies $dv/dt = 3t^3$. Find the position function (antiderivative).

$$\int 3x^3 dx$$

→ Integrate: $v(t) = \text{integral of } 3t^3 dt = 3/4t^4 + C$.

Answer: $\frac{3}{4}x^4 + C \Rightarrow \text{coeff} = 3/4$

